

Chapter 5

Operator Inequalities

5.1 Operator ordering

The set of Hermitian operators is equipped with a natural partial order, i.e., a consistent way of saying that one operator is larger than another one or that two operators are actually incomparable. In fact, we will come across various partial orders, but spend most time on what is sometimes called *Löwner's partial order* which is based on positive (semi-)definite differences.

First recall the basics about positive matrices. A matrix $A \in \mathcal{M}_d(\mathbb{C})$ is called *positive semi-definite* if $\langle \psi | A | \psi \rangle \geq 0$ for all $\psi \in \mathbb{C}^d$ and *positive definite* or *strictly positive* if the inequality is strict (i.e., ' > 0 ' holds). Being lazy we will occasionally just write 'positive' when meaning 'positive semi-definite'. This is equivalent to saying that A is Hermitian and has (strictly) positive eigenvalues and we will write $A \geq 0$ ($A > 0$). Recall that $A \geq 0$ is in turn equivalent to $A = B^\dagger B$ for some B and that $A \geq 0 \Rightarrow X^\dagger A X \geq 0$. An order relation for Hermitian matrices A, B is given by

$$A \geq B \quad \Leftrightarrow \quad A - B \geq 0. \quad (5.1)$$

This defines a partial order in the sense that two Hermitian matrices may be incomparable. If, for instance a Hermitian A has positive and negative eigenvalues, then neither $A \geq 0$ nor $A \leq 0$ holds. This is reminiscent to the ordering of events in Minkowski space, and for $\mathcal{M}_2(\mathbb{C})$ it is actually equivalent (see example 5.1).

Note that by the defining equation (5.1) a linear map $T : \mathcal{M}_d \rightarrow \mathcal{M}_{d'}$ is order preserving iff it is positive.

The above operator ordering leads to a peculiar structure: it makes the set of Hermitian matrices an *anti-lattice*. A partially ordered set is called a *lattice* if for every two elements, say A and B , there is an *infimum* (a largest lower bound, typically denoted $A \wedge B$) and a *supremum* (least upper bound, $A \vee B$). The set of Hermitian matrices in $\mathcal{M}_d$ ($d > 1$) is not a lattice (unlike the set of Hermitian projections; see example 5.2). It has the property that if two

elements happen to have a supremum or infimum, than this coincides with one of the elements—this structure is called an ‘anti-lattice’.

Before we come to the main operator inequalities involving quantum channels (or more generally positive maps) we collect some very useful basic results:

Theorem 5.1 (Douglas’ Theorem) ¹ *Let $A, B \in \mathcal{M}_d(\mathbb{C})$. Then the following are equivalent:*

1. $\text{range}(A) \subseteq \text{range}(B)$,
2. $AA^\dagger \leq \mu BB^\dagger$ for some $\mu \geq 0$,
3. $A = BC$ for some $C \in \mathcal{M}_d(\mathbb{C})$.

Moreover, if this is valid, then there is a unique C for which (i) $\ker(A) \subseteq \ker(C)$, (ii) $\text{range}(C) \subseteq \text{range}(B^\dagger)$ and (iii) $\|C\|_\infty^2 = \min\{\mu | AA^\dagger \leq \mu BB^\dagger\}$.

Theorem 5.2 (Block matrices and Schur complements) *Let $A \in \mathcal{M}_d(\mathbb{C})$ and $B \in \mathcal{M}_{d'}(\mathbb{C})$ be two positive semi-definite matrices and $C \in \mathcal{M}_{d,d'}(\mathbb{C})$. Then the following are equivalent:*

1. $\begin{pmatrix} A & C \\ C^\dagger & B \end{pmatrix} \geq 0$,
2. $\ker(B) \subseteq \ker(C)$ and $A \geq CB^{-1}C^\dagger$, where B^{-1} is the pseudo-inverse (i.e., inverse on the support) when necessary,
3. $\ker(B) \subseteq \ker(C)$ and $\|A^{-1/2}CB^{-1/2}\|_\infty \leq 1$.

Example 5.1 (Ordering in Minkowski space) *Every Hermitian 2×2 matrix can be expanded as $M = \sum_{i=0}^3 x_i \sigma_i$ with $x \in \mathbb{R}^4$. Since $\det(M) = x_0^2 - x_1^2 - x_2^2 - x_3^2 = \langle x, \eta x \rangle$ with $\eta := \text{diag}(1, -1, -1, -1)$ we can identify x with the coordinates of an event in Minkowski space, so that the determinant provides the Minkowski metric. As discussed in example 2.2 Lorentz transformation then take on the form $M \mapsto XMX^\dagger$ with $X \in SL(2, \mathbb{C})$.*

The ordering of events in Minkowski space, i.e., the causal structure imposed by the forward and backward light cones, then becomes the operator ordering on the level of the matrix representation. Recall that in Minkowski space two elements x and x' are time-like (i.e., potentially causally related) if $\langle x - x', \eta(x - x') \rangle$ is positive, and space-like (i.e., incomparable) if it is negative. The ordering between two time-like related events is then given by $(x - x')_0$, i.e., the sign of the time-component and x is in the forward/backward light cone of x' if this component is positive/negative.

On the level of the matrix representation $\det(M - M')$ tells us whether the matrices are comparable in matrix order and $M - M' \geq 0$ if and only iff $\det(M - M') \geq 0$ and $\text{tr}[M - M'] \geq 0$. In other words a Hermitian 2×2 matrix is positive iff its determinant and trace are.

In Minkowski space the absence of the lattice property means that for two space-like separated events there is no ‘latest’ event in the intersection of their backward light cones.

Example 5.2 (Lattice of projections) ...

¹which is typically referred to as ‘Douglas’ Lemma’.

5.2 Schwarz inequalities

Let $T : \mathcal{M}_d \rightarrow \mathcal{M}_{d'}$ be a completely positive, trace preserving map and T^* its dual (description in the Heisenberg picture). Then

$$\forall A \in \mathcal{M}_{d'} : \quad T^*(A^\dagger)T^*(A) \leq T^*(A^\dagger A), \quad (5.2)$$

is an operator version of the *Schwarz inequality*. Inequality (5.2) can easily be proven by employing Stinespring's representation $T^*(A) = V(A \otimes \mathbb{1})V^\dagger$ (with V an isometry, see Thm. 2.2) so that it can be written

$$V(A^\dagger \otimes \mathbb{1}) \underbrace{V^\dagger V}_{\leq \mathbb{1}} (A \otimes \mathbb{1}) V^\dagger \leq V(A^\dagger A \otimes \mathbb{1}) V^\dagger. \quad (5.3)$$

The following relaxes the assumptions on the map T and characterizes the cases of equality:

Theorem 5.3 (Operator Schwarz inequality) *Let $T^* : \mathcal{M}_{d'} \rightarrow \mathcal{M}_d$ be a positive linear map. If T is 2-positive, then for all $A, B \in \mathcal{M}_{d'}$:*

$$T^*(A^\dagger B)T^*(B^\dagger B)^{-1}T^*(B^\dagger A) \leq T^*(A^\dagger A), \quad (5.4)$$

where the inverse is taken on the range.

PROOF Define a rectangular matrix $C := (A, B)$ so that $C^\dagger C \in \mathcal{M}_{2d'}$ is positive semidefinite. Then, due to 2-positivity, $(\text{id}_2 \otimes T)(C^\dagger C) \geq 0$ which by Thm. 5.2 implies Eq. (5.4). $\square$

Theorem 5.4 (Equality in the operator Schwarz inequality) *Let $T^* : \mathcal{M}_{d'} \rightarrow \mathcal{M}_d$ be a positive linear map such that for a given $B \in \mathcal{M}_{d'}$ the Schwarz inequality in Eq. (5.4) holds for all $A \in \mathcal{M}_{d'}$. Denote by $\mathcal{A}_B \subseteq \mathcal{M}_{d'}$ the set of A 's for which equality is attained in Eq. (5.4) for the given B . Then for all $A \in \mathcal{A}_B$ we have*

$$T^*(A^\dagger B)T^*(B^\dagger B)^{-1}T^*(B^\dagger X) = T^*(A^\dagger X), \quad \forall X \in \mathcal{M}_{d'}. \quad (5.5)$$

PROOF Assume that $A \in \mathcal{A}_B$. Applying Eq. (5.4) (where we put everything to the r.h.s.) to B and $\tilde{A} = tA + X$ with $t \in \mathbb{R}$, we obtain

$$\begin{aligned} 0 \leq & t \left[T^*(A^\dagger X) + T^*(X^\dagger A) - T^*(A^\dagger B)T^*(B^\dagger B)^{-1}T^*(B^\dagger X) \right. \\ & \left. - T^*(X^\dagger B)T^*(B^\dagger B)^{-1}T^*(B^\dagger A) \right] \\ & + T^*(X^\dagger X) - T^*(X^\dagger B)T^*(B^\dagger B)^{-1}T^*(B^\dagger X), \end{aligned} \quad (5.6)$$

as the quadratic order in t vanishes due to $A \in \mathcal{A}_B$. In order for this inequality to hold for all $t \in \mathbb{R}$ the term linear in t has to vanish. By applying this argument to both X and iX and taking the sum we obtain Eq. (5.5). $\square$

Generalizations of the above inequalities to the case of block matrices can be easily obtained by setting $A = \sum_{i=1}^n A_i \otimes |i\rangle \in \mathcal{M}_{d',nd'}$ and $T = \tilde{T} \otimes \text{id}_n$ where $\tilde{T}$ has then to be $n+1$ -positive.

Positivity alone is not sufficient for a linear map to satisfy the Schwarz inequality (5.2); the simplest counterexample being matrix transposition with $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. However, there are notable exceptions:

Proposition 5.1 (Schwarz inequality for commutative domains) *Let $T : \mathcal{M}_d \rightarrow \mathcal{M}_{d'}$ be a positive linear map which is trace non-increasing (i.e., $T^*(\mathbb{1}) \leq \mathbb{1}$). Then the Schwarz-inequality (5.2) holds for all normal operators $A \in \mathcal{M}_{d'}$.²*

PROOF Take an arbitrary normal A . Then $[A, A^\dagger] = [A, A^\dagger A] = 0$ so that by Prop. 1.6 T^* is completely positive when restricted to the domain $\mathcal{D} := \text{span}\{A, A^\dagger, A^\dagger A, \mathbb{1}\}$. Following Prop. 1.7 we can extend this to a completely positive map on $\mathcal{M}_{d'}$ since $\mathcal{D}$ is an operator space. The assertion then follows from the fact that (i) the extension coincides with T^* on $\mathcal{D}$ and (ii) the Schwarz inequality holds for completely positive maps. $\square$

Prop. 5.1 is the basis of the following generalizations. For the first one recall that an operator $A \in \mathcal{B}(\mathcal{H})$ is called *subnormal* if there is a Hilbert space $\mathcal{H}' = \mathcal{H} \oplus \mathcal{H}^\perp$ and some B, C so that $\begin{pmatrix} A & B \\ 0 & C \end{pmatrix} \in \mathcal{B}(\mathcal{H}')$ is a normal operator. An alternative characterization of subnormal operators is that for all $n \in \mathbb{N}$ it holds that $\sum_{k,l=0}^n |l\rangle\langle k| \otimes A^{\dagger k} A^l \geq 0$ with $A^0 := \mathbb{1}$. An operator is called *quasinormal* iff $[A^\dagger A, A] = 0$ and the following relation between the mentioned sets holds:

$$\text{normal} \subset \text{quasinormal} \subset \text{subnormal}.$$

Theorem 5.5 (Schwarz inequality for subnormal operators) *Let $T : \mathcal{M}_d \rightarrow \mathcal{M}_{d'}$ be a positive linear map which is trace non-increasing (i.e., $T^*(\mathbb{1}) \leq \mathbb{1}$) and $A \in \mathcal{M}_{d'}$ a subnormal operator. Then*

$$T^*(A^\dagger)T^*(A) \leq T^*(A^\dagger A) \text{ and } T^*(A)T^*(A^\dagger) \leq T^*(A^\dagger A). \quad (5.7)$$

PROOF Denote by $N \in \mathcal{B}(\mathcal{H}')$ the normal extension of A and by $\theta : \mathcal{B}(\mathcal{H}') \rightarrow \mathcal{B}(\mathcal{H})$ the unital and completely positive linear map which maps $\theta(N) = A$. That is, θ returns the North-West block of its argument. Then, by Prop. 5.1

$$\tilde{T}(N^\dagger N) \geq \tilde{T}(N)\tilde{T}(N^\dagger), \tilde{T}(N^\dagger)\tilde{T}(N),$$

holds for the concatenated map $\tilde{T} := T^*\theta$ (and the second inequality follows from the first by $N \rightarrow N^\dagger$ and $NN^\dagger = N^\dagger N$). Writing things out in terms of T and A yields the claimed inequalities. $\square$

Another possibility of generalizing the Schwarz inequality to beyond 2-positive maps is replacing the $A^\dagger A$ term by something which is larger and commutes with A :

² A is *normal* if it commutes with its adjoint. So in particular Hermitian and unitary operators are normal.

Theorem 5.6 (Schwarz inequality for commuting dominant operators)

Let $T : \mathcal{M}_d \rightarrow \mathcal{M}_{d'}$ be a positive linear map which is trace non-increasing (i.e., $T^*(\mathbb{1}) \leq \mathbb{1}$) and $A \in \mathcal{M}_{d'}$. Then for every positive $D \in \mathcal{M}_{d'}$ which satisfies $[D, A] = 0$ and $D \geq A^\dagger A$ it holds that

$$T^*(A^\dagger)T^*(A) \leq T^*(D) \text{ and } T^*(A)T^*(A^\dagger) \leq T^*(D). \quad (5.8)$$

PROOF First note that $D \geq AA^\dagger$. In order to see this, assume for the moment that D is invertible. Then $D \geq A^\dagger A$ implies that $\mathbb{1} \geq X^\dagger X$ with $X := AD^{-1/2}$. This in turn implies $\mathbb{1} \geq XX^\dagger$ from which $D \geq AA^\dagger$ follows by using $[D, A] = 0$. This generalizes to non-invertible D 's if we first replace $D \rightarrow D + \epsilon\mathbb{1}$ in the argument and then let $\epsilon \rightarrow 0$.

Now construct the block matrix

$$N := \begin{pmatrix} A & \sqrt{D - AA^\dagger} \\ \sqrt{D - A^\dagger A} & -A^\dagger \end{pmatrix}.$$

Direct calculation (and using that $A\sqrt{D - A^\dagger A} = \sqrt{D - AA^\dagger}A$) shows that N is a normal operator. Analogous to the proof of Thm.5.5 we can now introduce a unital and completely positive linear map $\theta : N \rightarrow A$ and show the desired inequalities by using Prop.5.1 in order to get $\tilde{T}(N^\dagger N) \geq \tilde{T}(N^\dagger)\tilde{T}(N)$, $\tilde{T}(N)\tilde{T}(N^\dagger)$ for the map $\tilde{T} := T^*\theta$. Using that $\theta(N^\dagger N) = D$ and $\theta(N) = A$ concludes the proof. $\square$

One might wonder whether a unital positive map T^* which satisfies the Schwarz inequality in Eq.(5.2) is automatically 2-positive. The following example shows that this is not the case. Since the set of unital maps which satisfy Eq.(5.2) is therefore larger than the subset of 2-positive maps but smaller than the set of positive maps it may deserve an own name: *Schwarz maps* is the most commonly used; $\frac{3}{2}$ -positive maps a not too serious alternative.

Example 5.3 (Schwarz maps which are not 2-positive) Consider the map $T : \mathcal{M}_2 \rightarrow \mathcal{M}_2$ defined via

$$T^*(A) := \frac{1}{2} \left(A^T + \frac{1}{2} \text{tr}[A] \mathbb{1} \right). \quad (5.9)$$

As a convex combination of two positive maps it is positive. However, it fails to be 2-positive (which in this case means completely positive) since the Choi-Jamiolkowski operator of T^* is $(\mathcal{F} + \frac{1}{2}\mathbb{1})/4$ which has one eigenvalue $-1/8$. Nevertheless, the map satisfies the Schwarz inequality in Eq.(5.2). In order to see this define $F(A) := T^*(A^\dagger A) - T^*(A^\dagger)T^*(A)$ and note that $F(A) = F(A + \lambda\mathbb{1})$ for all $\lambda \in \mathbb{C}$ and $A \in \mathcal{M}_2$. So for proving that $F(A) \geq 0$ for all A we can w.l.o.g. assume that $\text{tr}[A] = 0$. In this case

$$\begin{aligned} F(A) &= \frac{1}{4} \left(2(A^\dagger A)^T + \text{tr}[A^\dagger A] \mathbb{1} - \bar{A}A^T \right) \\ &\geq \frac{1}{2} (A^\dagger A)^T \geq 0, \end{aligned} \quad (5.10)$$

since $\text{tr}[A^\dagger A] \mathbb{1} = \text{tr}[\bar{A}A^T] \mathbb{1} \geq \bar{A}A^T$.

By Thm.5.3 every 2-positive linear map T which is trace non-increasing (i.e., $T^*(\mathbb{1}) \leq \mathbb{1}$) satisfies the Schwarz inequality (5.2). The cases of equality in this inequality lead us to the sets

$$\mathcal{A}^R := \{A \in \mathcal{M}_{d'} \mid T^*(XA) = T^*(X)T^*(A), \forall X \in \mathcal{M}_{d'}\}, \quad (5.11)$$

$$\mathcal{A}^L := \{A \in \mathcal{M}_{d'} \mid T^*(AX) = T^*(A)T^*(X), \forall X \in \mathcal{M}_{d'}\}, \quad (5.12)$$

called the *right/left multiplicative domain* of T^* , and $\mathcal{A} := \mathcal{A}^R \cap \mathcal{A}^L$ its *multiplicative domain*.

Note that these three sets form each an algebra (i.e., they are in particular closed under multiplication) and that T^* acts as a homomorphism on these algebras. Since the multiplicative domain $\mathcal{A}$ is in addition closed under taking the adjoint, it is a $*$ -algebra and T^* a $*$ -homomorphism when restricted to $\mathcal{A}$.

Theorem 5.7 (Multiplicative domains) *Let T be a linear map which fulfills the Schwarz inequality (5.2). Then*

$$\{A \in \mathcal{M}_{d'} \mid T^*(A^\dagger A) = T^*(A^\dagger)T^*(A)\} = \mathcal{A}^R, \quad (5.13)$$

$$\{A \in \mathcal{M}_{d'} \mid T^*(AA^\dagger) = T^*(A)T^*(A^\dagger)\} = \mathcal{A}^L. \quad (5.14)$$

The proof of this statement is analogous to that of Thm.5.4. The fact that the set of operators achieving equality in the Schwarz inequality forms an algebra has remarkable consequences when it comes to the discussion of fixed points of quantum channels.

5.3 Operator convexity and monotonicity

Functional calculus gives a meaning to scalar functions evaluated on operators. Suppose a matrix $A \in \mathcal{M}_d(\mathbb{C})$ can be diagonalized as $A = UDU^{-1}$ and a given function f is defined on a domain including the eigenvalues of A (i.e., the entries of the diagonal matrix D). Then a consistent way of defining f applied to A is

$$f(A) = U f(D) U^{-1} \quad \text{where} \quad [f(D)]_{ii} = f(D_{ii}). \quad (5.15)$$

As diagonalizable matrices are dense, there is at most one continuous way of extending the definition of $f(A)$ to all $A \in \mathcal{M}_d(\mathbb{C})$. The core of this is the observation that for diagonalizable matrices Eq.(5.15) implies that $f(A) = g(A)$ for every polynomial g which coincides with f on the spectrum of A . In general one can show that replacing f by a such a polynomial g gives a well-defined meaning to $f(A)$ if the derivatives of f and g coincide up to order $j-1$ at every eigenvalue of A with Jordan-block of size j .

As we are only considering Hermitian matrices in this chapter we can, however, use Eq.(5.15) and take $U^{-1} = U^\dagger$ unitary. The above definition implies that $f(VAV^\dagger) = Vf(A)V^\dagger$ for every unitary V . Moreover, $f(A \oplus B) = f(A) \oplus f(B)$ and in particular $f(A \otimes \mathbb{1}) = f(A) \otimes \mathbb{1}$. Be aware, that in general $f(A \otimes B) \neq f(A) \otimes f(B)$ and $f(VAV^\dagger) \neq Vf(A)V^\dagger$ if V is merely an isometry.

Having defined functions on operators and Löwner's operator ordering in mind we may now ask how properties of functions like monotonicity or convexity translate to the world of matrices and operators.

Operator monotonicity and operator convexity . Recall that a real-valued function f is called *convex* on an interval $I \subseteq \mathbb{R}$ if for all $\lambda \in [0, 1]$ and all $a, b \in I$ we have

$$f(\lambda a + (1 - \lambda)b) \leq \lambda f(a) + (1 - \lambda)f(b), \quad (5.16)$$

i.e., the second derivative (if it exists) should be positive on I . For continuous functions this is equivalent to *mid-point convexity*, which is the same statement with $\lambda = 1/2$. f is called *concave* if $-f$ is convex, and it is called *monotone* on I if for all $a, b \in I$

$$a \geq b \Rightarrow f(a) \geq f(b). \quad (5.17)$$

If Eqs.(5.16,5.17) hold for all Hermitian matrices $A, B \in \mathcal{M}_d(\mathbb{C})$ whose spectrum is contained in I , then we say that f is *matrix convex* or *matrix monotone* of order d on I . Functions which are convex/monotone on scalars need not be so on matrices (see example 5.4). In fact, the set of matrix convex/monotone functions becomes smaller and smaller when we increase d . Functions which remain matrix convex/monotone for all $d \in \mathbb{N}$ are called *operator convex* or *operator monotone*, respectively.

Note that the sets of operator monotone functions and operator convex functions (on a given interval) are both convex cones which are closed under pointwise limits. So, characterizing all such functions essentially boils down to identifying the extreme rays of these sets and then allowing for positive linear combinations. The two following theorems do this for the positive half-line (for other intervals analogous results hold).

Theorem 5.8 (Operator convexity) *A function $f : [0, \infty) \rightarrow \mathbb{R}$ is operator convex iff it can be written as*

$$f(x) = a + bx + cx^2 + \int_0^\infty \frac{yx^2}{y+x} d\mu(y), \quad (5.18)$$

where $a, b \in \mathbb{R}$, $c \geq 0$ and μ is a positive finite measure.

Theorem 5.9 (Löwner's theorem—operator monotonicity) *For a continuous function $f : (0, \infty) \rightarrow \mathbb{R}$ the following statements are equivalent:*

1. f is operator monotone,
2. f has an analytic continuation to the entire domain $\text{Im} z \neq 0$ such that it maps the upper half-plane $\{z \in \mathbb{C} | \text{Im} z > 0\}$ into itself,
3. f has an integral representation of the form

$$f(x) = a + bx + \int_{-\infty}^0 \frac{1+xy}{y-x} d\mu(y), \quad (5.19)$$

where $a \in \mathbb{R}$, $\beta \geq 0$ and μ is a finite positive measure.

In particular the relation between operator monotone functions and so-called *Pick functions* (holomorphic functions with positive imaginary part) is a very powerful result. It enables us to ‘see’ right away that certain functions (like the exponential function) are not operator monotone. The integral representations of operator convex and operator monotone functions seem somehow related and, in fact, there are close relations between these two classes. For instance, a continuous function $f : [0, \infty) \rightarrow [0, \infty)$ is operator monotone iff it is operator concave.

Example 5.4 (Operator monotone/convex functions) *The following lists some functions (in the variable $x \in \mathbb{R}$) and their properties regarding operator convexity/monotonicity on specified intervals $I \subseteq \mathbb{R}$:*

<i>operator monotone</i>	<i>operator convex</i>
◦ $a + bx$ on $\mathbb{R}$ for $a \in \mathbb{R}, b \geq 0$ ◦ x^p on $[0, \infty)$ for $p \in [0, 1]$ ◦ $-x^p$ on $(0, \infty)$ for $p \in [-1, 0]$ ◦ $\log x$ on $(0, \infty)$ ◦ $(x \log x)/(x - 1)$ on $(0, \infty)$ ◦ $\frac{x}{y+x}$ on $(0, \infty)$ for $y \geq 0$ ◦ $\tan x$ on $(-\pi/2, \pi/2)$	◦ $a + bx + cx^2$ on $\mathbb{R}$ for $a, b \in \mathbb{R}, c \geq 0$ ◦ x^p on $(0, \infty)$ for $p \in [1, 2]$ ◦ x^p on $(0, \infty)$ for $p \in [-1, 0]$ ◦ $-\log x$ on $(0, \infty)$ ◦ $x \log x$ on $(0, \infty)$ ◦ $\frac{x^2}{y+x}$ on $[0, \infty)$ for $y \geq 0$ ◦ $f : (0, \infty) \rightarrow (0, \infty)$ if $x \mapsto 1/f(x)$ is op. monotone
<i>not operator monotone</i>	<i>not operator convex</i>
◦ x^p on $(0, \infty)$ for $p \notin [0, 1]$ ◦ e^x on any non-trivial I ◦ $\max(0, x)$ on any $I \supset \{0\}$	◦ x^p on $(0, \infty)$ for $p \notin [-1, 0] \cup [1, 2]$ ◦ e^x on any non-trivial I ◦ x on any $I \supset \{0\}$

For ‘simple’ functions which are not operator monotone/convex there are similarly ‘simple’ counterexamples. Take for instance

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \quad (5.20)$$

When applied to them we see that $f(x) = x^2$ is not operator monotone on $[0, \infty)$ and that $f(x) = x^3$ fails to be operator convex on $[0, \infty)$.

We will now connect first operator convexity and then operator monotonicity to positive maps. Recall that positive maps are exactly the ones which preserve the order, i.e., if $A \geq B$, then $T(A) \geq T(B)$ for positive T . Conversely, if this holds for all A, B , then T has to be positive.

A typical requirement in the context addressed below is that the positive map satisfies $T(\mathbb{1}) \leq \mathbb{1}$. This implies that T is contractive in the following sense: consider a Hermitian A with $\text{spec}(A) \subseteq [a, b]$, where $[a, b]$ is an interval containing zero. Then $T(\mathbb{1}) \leq \mathbb{1}$ implies that also $\text{spec}(T(A)) \subseteq [a, b]$ since

$$a\mathbb{1} \leq T(a\mathbb{1}) \leq T(A) \leq T(b\mathbb{1}) \leq b\mathbb{1}. \quad (5.21)$$

Theorem 5.10 (Operator convexity from projection inequality) *Consider a real-valued function f defined on some interval $I \subseteq \mathbb{R}$ with $0 \in I$. If for all*

$d \in \mathbb{N}$, all Hermitian projections $P \in \mathcal{M}_d(\mathbb{C})$ and all Hermitian $A \in \mathcal{M}_d(\mathbb{C})$ with $\text{spec}(A) \subset I$

$$f(PAP) \leq Pf(A)P, \quad \text{then} \quad (5.22)$$

1. $f(0) \leq 0$,
2. f is operator convex on I ,
3. for all positive linear maps $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C})$ with $T(\mathbb{1}) \leq \mathbb{1}$ and all Hermitian A with spectrum in I :

$$f(T(A)) \leq T(f(A)). \quad (5.23)$$

PROOF 1. follows from Eq.(5.22) already by setting $A = P^\perp := \mathbb{1} - P$. In order to show 3. consider first a completely positive and unital map T . Using Stinespring's theorem (Thm.2.2) and expressing the isometry as a unitary times a projection we can write

$$T(A) \oplus 0_{dr-d'} = PU^\dagger(A \otimes \mathbb{1}_r)UP. \quad (5.24)$$

When applying f to both sides of this inequality we can use $f(U^\dagger \cdot U) = U^\dagger f(\cdot)U$, and $f(A \otimes \mathbb{1}) = f(A) \otimes \mathbb{1}$ so that the Eq.(5.23) indeed follows from (5.22) for unital and completely positive T . Now let us successively get rid of the additional constraints. Assume T is not unital. Then we can define a unital map $T' : \mathcal{M}_d(\mathbb{C}) \oplus \mathbb{C} \rightarrow \mathcal{M}_{d'}(\mathbb{C})$ by $T' : A \oplus b \mapsto T(A) + b(\mathbb{1} - T(\mathbb{1}))$ (and off-diagonals are mapped to zero). T' is completely positive iff T is completely positive and $T(\mathbb{1}) \leq \mathbb{1}$. Applying Eq.(5.23) to T' then yields the same inequality for T , so unitality can be replaced by $T(\mathbb{1}) \leq \mathbb{1}$. If T is merely positive we can use that A and $f(A)$ are elements of an abelian C^* -algebra (the one generated by A and $\mathbb{1}$) on which T is completely positive (Prop.1.6) with a completely positive extension on $\mathcal{M}_d$ (Prop.1.7). This completes the proof of 3.

Finally, 2. follows by applying Eq.(5.23) to the map $T : A \oplus B \rightarrow \lambda A + (1 - \lambda)B$, $\lambda \in [0, 1]$. $\square$

This theorem becomes yet more interesting as the converse also holds:

Theorem 5.11 (Projection inequality from operator convexity) *Let $f : I \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be an operator convex function on an interval $I \ni 0$. Then for all Hermitian projections $P \in \mathcal{M}_d(\mathbb{C})$ (with $P^\perp := \mathbb{1} - P$) and all Hermitian $A \in \mathcal{M}_d(\mathbb{C})$ with $\text{spec}(A) \subset I$:*

$$f(PAP) \leq Pf(A)P + P^\perp f(0)P^\perp. \quad (5.25)$$

Consequently, if $f(0) \leq 0$, then for all positive maps T with $T(\mathbb{1}) \leq \mathbb{1}$:

$$f(T(A)) \leq T(f(A)). \quad (5.26)$$

PROOF Define a unitary $U := P - P^\perp$. Then

$$P f(PAP) P = P f(PAP + P^\perp A P^\perp) P \quad (5.27)$$

$$= P f(A/2 + UAU^\dagger/2) P \quad (5.28)$$

$$\leq P \left(f(A)/2 + U f(A) U^\dagger/2 \right) P = P f(A) P, \quad (5.29)$$

where the inequality is the assumed operator convexity. Eq.(5.25) then follows from $f(PAP) = P f(PAP) P + P^\perp f(0) P^\perp$ and Eq.(5.26) is just Eq.(5.23) since Thm.5.10 applies if $f(0) \leq 0$. $\square$

The following shows that we can drop the restriction $f(0) \leq 0$ if T is normalized, i.e., unital:

Theorem 5.12 (Operator convexity and unital positive maps) *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be operator convex on $[0, \infty)$ and $T : \mathcal{M}_d \rightarrow \mathcal{M}_{d'}$ any positive unital map. Then for all $A \geq 0$:*

$$f(T(A)) \leq T(f(A)). \quad (5.30)$$

PROOF By the integral representation in Thm.5.8 it suffices to prove Eq.(5.30) for $f(x) = x^2$ and $f(x) = x^2/(y+x)$, $y > 0$. The first case is nothing but the Schwarz inequality in Eq.(5.2). The second case can be simplified by rewriting it as $f(x) = x - y + y^2/(x+y)$. Inserting this into Eq.(5.30) and exploiting linearity and unitality of T we see that proving Eq.(5.30) boils down to showing that

$$T(B)^{-1} \leq T(B^{-1}), \quad (5.31)$$

for all $B > 0$. In order to show Eq.(5.31) note that by introducing the spectral decomposition $B = \sum_i b_i P_i$ (with spectral projections P_i) and using that $\mathbb{1} = T(\mathbb{1}) = \sum_i T(P_i)$ we get

$$\begin{pmatrix} T(B^{-1}) & \mathbb{1} \\ \mathbb{1} & T(B) \end{pmatrix} = \sum_i \begin{pmatrix} b_i^{-1} & 1 \\ 1 & b_i \end{pmatrix} \otimes T(P_i) \geq 0. \quad (5.32)$$

From here Eq.(5.31) follows by invoking the condition for the positivity of block matrices in Thm.5.2. $\square$

By applying this to $f(x) = x \log x$, which is operator convex on the positive half-line, we obtain:

Corollary 5.1 (Entropic operator inequality) *Let $T : \mathcal{M}_d \rightarrow \mathcal{M}_{d'}$ be a positive unital map. Then for all $\rho \geq 0$:*

$$T(\rho) \log T(\rho) \leq T(\rho \log \rho). \quad (5.33)$$

Problem 10 *Provide a generalization of the inequality for positive maps with $T(\leq \mathbb{1})$ in the form of Eq.(5.30) for arbitrary operator convex f (not necessarily satisfying $f(0) \leq 0$).*

Let us now turn from convexity to monotonicity:

Theorem 5.13 (Operator monotonicity and positive maps) *Let f be an operator monotone function on an interval $I = [0, a]$ with $f(0) \geq 0$ and $T : \mathcal{M}_d \rightarrow \mathcal{M}_{d'}$ any positive map for which $T(\mathbb{1}) \leq \mathbb{1}$. Then for all $A = A^\dagger$ with $\text{spec}(A) \in I$:*

$$T(f(A)) \leq f(T(A)). \quad (5.34)$$

PROOF Let us first prove the statement for $T(A) = X^\dagger A X$ with $\|X\|_\infty \leq 1$. To this end define a unitary block matrix

$$U := \begin{pmatrix} X & \sqrt{\mathbb{1} - X X^\dagger} \\ \sqrt{\mathbb{1} - X^\dagger X} & -X^\dagger \end{pmatrix}, \quad (5.35)$$

and note that the upper left block of $U^\dagger(A \oplus 0)U$ is $X^\dagger A X$. Hence by Thm.5.2 we can find for every $\epsilon > 0$ a $\delta > 0$ such that $U^\dagger(A \oplus 0)U \leq (X^\dagger A X + \epsilon \mathbb{1}) \oplus (\delta \mathbb{1})$. Applying f to this inequality, using its monotonicity and that $f(U^\dagger \cdot U) = U^\dagger f(\cdot) U$ we get

$$X^\dagger f(A) X + \sqrt{\mathbb{1} - X X^\dagger} f(0) \sqrt{\mathbb{1} - X X^\dagger} \leq f(X^\dagger A X + \epsilon \mathbb{1}) \quad (5.36)$$

by comparing the upper left blocks on both sides. The asserted inequality (5.34) follows then from continuity of operator monotone functions by taking $\epsilon \rightarrow 0$ together with $f(0) \geq 0$.

This can be extended to all completely positive maps with $T(\mathbb{1}) \leq \mathbb{1}$ by Stinespring's theorem. Following Thms.2.2,2.3 we can represent every such map in the form $T(A) = X^\dagger(A \otimes \mathbb{1})X$, where $X^\dagger X = T(\mathbb{1}) \leq \mathbb{1}$. By invoking $f(A \otimes \mathbb{1}) = f(A) \otimes \mathbb{1}$ we can thus generalize the inequality to contractive completely positive maps. Finally, if T is merely positive with $T(\mathbb{1}) \leq \mathbb{1}$, then it is completely positive on the abelian algebra generated by $\mathbb{1}$ and A (Prop.1.6) and admits a completely positive extension on $\mathcal{M}_d$ (Prop.1.7). $\square$

Applying this to some of the examples in example 5.4 leads to:

Corollary 5.2 *Let $T : \mathcal{M}_d \rightarrow \mathcal{M}_{d'}$ be a positive map with $T(\mathbb{1}) \leq \mathbb{1}$. Then*

1. $T(A) \leq T(A^p)^{1/p}$ for all $p \geq 1$ and $A \geq 0$,
2. $T(A) \geq T(A^p)^{1/p}$ for all $A > 0$ and³ $p \in [1/2, 1]$,
3. $T(\log(A)) \leq \log(T(A))$ for all $A > 0$.

Problem 11 *Prove a counterpart of Thm.5.12 for matrix convex functions of finite order. In other words, find a proof which does not use the integral representation (or any other representation implying operator convexity).*

Let us finally prove a useful implication of the above theorems:

Corollary 5.3 *Let $\{T_i : \mathcal{M}_d \rightarrow \mathcal{M}_{d'}\}_{i=1,\dots,n}$ be a collection of positive linear maps such that $\sum_i T_i(\mathbb{1}) \leq \mathbb{1}$.*

³This follows from Thm.5.11 and operator convexity of $x \mapsto x^{1/p}$ for $p \in [1/2, 1]$.

1. If f is operator monotone on an interval $I \ni 0$ with $f(0) \geq 0$, then for all Hermitian operators A_i with spectrum in I :

$$\sum_{i=1}^n T_i(f(A_i)) \leq f\left(\sum_{i=1}^n T_i(A_i)\right). \quad (5.37)$$

2. If f is operator convex on an interval $I \ni 0$ and $f(0) \leq 0$, then for all Hermitian operators A_i with spectrum in I :

$$\sum_{i=1}^n T_i(f(A_i)) \geq f\left(\sum_{i=1}^n T_i(A_i)\right). \quad (5.38)$$

PROOF Let us construct a new map $T : \mathcal{M}_d \otimes \mathcal{M}_n \rightarrow \mathcal{M}_{d'}$ via $T(A) := \sum_{i=1}^n T_i(V_i A V_i^\dagger)$ with isometries $V_i := \mathbb{1}_d \otimes |i\rangle$. Similarly, define $A := \sum_i A_i \otimes |i\rangle\langle i|$. Then $T(A) = \sum_i T_i(A_i)$ and $f(A) = \sum_i f(A_i) \otimes |i\rangle\langle i|$. Since T is by construction such that $T(\mathbb{1}) \leq \mathbb{1}$ we can now apply Thm.5.11 and Thm.5.13 leading to assertions 2. and 1., respectively. $\square$

5.4 Joint convexity

Joint convexity is a property of certain functions of several variables. A function $f : \mathcal{D} \subseteq \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ with domain $\mathcal{D}$ is called *jointly convex* in its arguments iff

$$f(a, b) \leq \lambda f(a_1, b_1) + (1 - \lambda) f(a_2, b_2) \quad (5.39)$$

holds for all a_i, b_i with $a = \lambda a_1 + (1 - \lambda) a_2$, $b = \lambda b_1 + (1 - \lambda) b_2$ and all $\lambda \in [0, 1]$. It is defined analogously for more than two arguments and f is called *jointly concave* iff $-f$ is jointly convex. Clearly, a jointly convex function is convex in each of its arguments separately (i.e., $f(a, b) \leq \lambda f(a_1, b) + (1 - \lambda) f(a_2, b)$); the converse is, however, not true in general.

In the following we will apply the concept of joint convexity to operators, i.e., we will essentially replace $\mathbb{R}$ with the space of Hermitian operators. Two related types of joint convexity results will be of interest: (i) joint operator convexity, where the ordering " $\leq$ " in Eq.(5.39) refers to the partial order in the space of Hermitian operators (see Eq.(5.1)) and f is a mapping between operators, and (ii) joint convexity of functionals which map operators into $\mathbb{R}$ (in most cases the 'functional' will essentially be the trace).

Joint operator convexity We consider mappings of the form $F : \mathcal{D} \subseteq \mathcal{M}_{d_1} \times \cdots \times \mathcal{M}_{d_n} \rightarrow \mathcal{M}_d$ and say that they are *jointly operator convex* in the domain $\mathcal{D}$ iff

$$F(A) \leq \sum_i \lambda_i F(A^{(i)}), \quad (5.40)$$

for all $A, A^{(i)} \in \mathcal{D}$ which are related via $A = \sum_i \lambda_i A^{(i)}$ with λ being a vector of probabilities. The ordering ' $\leq$ ' used in this context is the one between Hermitian operators which means in particular that F should have Hermitian image.

The following theorem encompasses most of the results about joint operator convexity. It provides a tool for constructing maps in several variables from maps in fewer variables while retaining the convexity properties. In particular, it allows us to construct mappings which are jointly operator concave/convex from simple operator concave/convex functions:

Theorem 5.14 (Joint operator convexity) *Let $g : \mathcal{D} \subseteq \mathcal{M}_{d_1} \times \cdots \times \mathcal{M}_{d_n} \rightarrow \mathcal{M}_d$ be a map on the direct product $\mathcal{D}$ of n positive operators, and similarly $h : \mathcal{D}' \subseteq \mathcal{M}_{d'_1} \times \cdots \times \mathcal{M}_{d'_m} \rightarrow \mathcal{M}_d$. Assume that g is jointly operator concave and positive and h is semi-definite. Let $I \ni 0$ be the positive/negative real half line depending on whether h is positive or negative semi-definite. For any function $f : I \rightarrow \mathbb{R}$ with $f(0) \leq 0$ define $F : \mathcal{D}' \times \mathcal{D} \rightarrow \mathcal{M}_d$ as*

$$F(L, R) := \sqrt{g(R)} f\left(g(R)^{-1/2} h(L) g(R)^{-1/2}\right) \sqrt{g(R)}. \quad (5.41)$$

We consider joint operator convexity of F in its $n + m$ arguments. F is jointly operator convex on positive operators for which g is invertible if at least one of the following holds:

1. h is jointly operator concave and f is operator anti-monotone,
2. h is affine and f is operator convex.

PROOF Consider the convex decompositions $R = \sum_i \lambda_i R_i$ and $L = \sum_i \lambda_i L_i$ and define $K_i := \sqrt{\lambda_i g(R_i)} g(R)^{-1/2}$. This satisfies

$$\sum_i K_i^\dagger K_i = g(R)^{-1/2} \left(\sum_i \lambda_i g(R_i) \right) g(R)^{-1/2} \leq \mathbb{1}, \quad (5.42)$$

where the inequality comes from joint concavity of g . Assume that the requirements of the first assertion are fulfilled. Then if we insert the definitions:

$$F(L, R) \leq \sqrt{g(R)} f\left(\sum_i \lambda_i g(R)^{-1/2} h(L_i) g(R)^{-1/2}\right) \sqrt{g(R)}, \quad (5.43)$$

$$= \sqrt{g(R)} f\left(\sum_i K_i^\dagger g(R_i)^{-1/2} h(L_i) g(R_i)^{-1/2} K_i\right) \sqrt{g(R)}, \quad (5.44)$$

$$\leq \sqrt{g(R)} \sum_i K_i^\dagger f\left(g(R_i)^{-1/2} h(L_i) g(R_i)^{-1/2}\right) K_i \sqrt{g(R)}, \quad (5.45)$$

$$= \sum_i \lambda_i F(L_i, R_i), \quad (5.46)$$

where the first inequality comes from concavity of h together with anti-monotonicity of f and the second inequality follows from the first part of Cor.5.3 by setting $T_i(\cdot) := K_i^\dagger \cdot K_i$ and noticing that Eq.(5.42) means $\sum_i T_i(\mathbb{1}) \leq \mathbb{1}$. For the second assertion of the theorem note that if h is affine we have equality in Eq.(5.43). The inequality in Eq.(5.45) follows then from the second part of Cor.5.3. $\square$

Note that in the preceding theorem we may extend $\mathcal{D}'$ to all Hermitian operators and allow h to have Hermitian image if we set $I = \mathbb{R}$. Similarly, we can

enlarge $\mathcal{D}$ as long as g has positive image. Applying Thm.5.14 to various functions now provides us with a zoo of jointly operator convex/concave mappings:

- Corollary 5.4** 1. $\mathcal{M}_d \times \mathcal{M}_d \ni (L, R) \mapsto LR^{-\alpha}L$ is jointly operator convex for $\alpha \in [0, 1]$ on positive L and Hermitian R ,
2. $\mathcal{M}_d \times \mathcal{M}_d \ni (L, R) \mapsto R^{\beta/2}(R^{-\beta/2}L^\gamma R^{-\beta/2})^\alpha R^{\beta/2}$ is jointly operator concave on positive operators for $\alpha, \beta, \gamma \in [0, 1]$,
3. $\mathcal{M}_d^m \times \mathcal{M}_d \ni (L, R) \mapsto R^{\beta/2}f(\sum_{i=1}^m R^{-\beta/2}L_i R^{-\beta/2})R^{\beta/2}$, $\beta \in [0, 1]$ is jointly operator convex on positive operators if f is operator convex with $f(0) \leq 0$, e.g. $f(x) = -x^\alpha$, $f(x) = x^{1+\alpha}$ with $\alpha \in [0, 1]$ or $f(x) = x \log x$,

PROOF Use Thm.5.14 and for 1. set $f(x) = x^2$, $h(x) = x$ and $g(x) = x^\alpha$. For 2. set $f(x) = -x^\alpha$, $g(x) = x^\beta$, $h(x) = x^\gamma$. For 3. set $h(L) = \sum_i L_i$ and $g(x) = x^\beta$. $\square$

By replacing L, R in the above discussion by $L \rightarrow L \otimes \mathbb{1}$ and $R \rightarrow \mathbb{1} \otimes R$ and thereby making things commutative, we obtain similar statements for tensor product mappings:

- Corollary 5.5** 1. $\mathcal{M}_d \times \mathcal{M}_d \ni (L, R) \mapsto L^x \otimes R^y$ is jointly operator concave on positive operators for $x, y \geq 0$ with $x + y \leq 1$.
2. $\mathcal{M}_d^m \ni (A_1, \dots, A_m) \mapsto \bigotimes A_i^{x_i}$ is jointly operator concave on positive operators for $x_i \geq 0$ with $\sum_i x_i \leq 1$.
3. $\mathcal{M}_d \times \mathcal{M}_d \ni (L, R) \mapsto L^{1+\alpha} \otimes R^{-\beta\alpha}$ is jointly operator convex on positive operators for $\alpha, \beta \in [0, 1]$.

PROOF Statement 1. follows by inserting into the second assertion of Cor.5.4. Statement 2. follows by induction from the previous argument. That is, for $m = 3$ we continue with $h(A_1, A_2) = A_1^x \otimes A_2^x \otimes \mathbb{1}$, exploit that it is concave by the first statement and apply Thm.5.14 with $R = \mathbb{1} \otimes \mathbb{1} \otimes A_3$ and $f(x) = -x^\alpha$ and $g(x) = x^\beta$, $\alpha, \beta \in [0, 1]$, etc. Assertion 3. follows from point 3. of Cor.5.4 when applied to $f(x) = x^{1+\alpha}$. $\square$

Jointly convex functionals In the following we will discuss joint convexity/concavity properties of functionals, i.e., maps from a set of operators into $\mathbb{R}$. In most cases the results follow from the previous ones on joint operator convexity by taking the trace or the expectation value w.r.t. some positive operator. One of the central results in this context is the following:

Theorem 5.15 (Ando-Lieb) For every $K \in \mathcal{M}_d(\mathbb{C})$ the map

$$\mathcal{M}_d(\mathbb{C}) \times \mathcal{M}_d(\mathbb{C}) \ni (A, B) \mapsto \text{tr}[K^\dagger A^x K B^y] \quad (5.47)$$

- is jointly concave on positive operators for $x, y \geq 0$ if $x + y \leq 1$, and

- jointly convex on positive operators for $x \in [1, 2]$ if $y \in [1 - x, 0]$.

PROOF The result is an immediate consequence of items 1. and 3. of Cor. 5.5. In order to see this consider the expectation value of $L^x \otimes R^y$ w.r.t. $|\psi\rangle = (K \otimes \mathbb{1}) \sum_{i=1}^d |ii\rangle$. Using Eq. (1.12) this gives

$$\langle \psi | L^x \otimes R^y | \psi \rangle = \text{tr} [K^\dagger L^x K (R^T)^y], \quad (5.48)$$

so that the result follows by setting $A = L$ and $B = R^T$. $\square$

In fact, the parameter ranges for x and y in Thm. 5.15 are the only ones (apart from interchanging $x \leftrightarrow y$) for which joint convexity/concavity holds.

Theorem 5.16 (Monotonicity under cp-maps) *Consider a functional $F : \mathcal{D} \subseteq \mathcal{M}_d \times \cdots \times \mathcal{M}_d \rightarrow \mathbb{R}$ which is defined for all dimensions $d \in \mathbb{N}$. Assume that F satisfies*

1. Joint convexity in $\mathcal{D}$,
2. Unitary invariance, i.e., for all $A \in \mathcal{D}$ and all unitaries $U \in \mathcal{M}_d(\mathbb{C})$ it holds that $F(UA_1U^\dagger, \dots, UA_nU^\dagger) = F(A_1, \dots, A_n)$,
3. Invariance under tensor products, meaning that for all $A \in \mathcal{D}$ and all density operators $\tau \in \mathcal{M}_{d'}(\mathbb{C})$ we have $F(A_1 \otimes \tau, \dots, A_n \otimes \tau) = F(A_1, \dots, A_n)$.

Then F is monotone w.r.t. all completely positive trace-preserving maps $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C})$ in the sense that for all $A \in \mathcal{D}$

$$F(T(A_1), \dots, T(A_n)) \leq F(A_1, \dots, A_n). \quad (5.49)$$

PROOF Let us follow Thm. 2.5 and represent T as $T(\rho) = \text{tr}_E [U(\rho \otimes \tau)U^\dagger]$ where τ is a density matrix, U a unitary and tr_E the partial trace over an ‘environmental’ system of dimension m , say. If we take a unitary operator basis $\{V_i\}_{i=1, \dots, m^2}$ in $\mathcal{M}_m$ (orthonormal w.r.t. the Hilbert-Schmidt inner product—see Exp. 2.1), we can write

$$T(\rho) \otimes \mathbb{1}_m / m = \frac{1}{m^2} \sum_{i=1}^{m^2} (\mathbb{1} \otimes V_i) U(\rho \otimes \tau) U^\dagger (\mathbb{1} \otimes V_i^\dagger). \quad (5.50)$$

Observing that $\mathbb{1}_m / m$ is again a density matrix and that the r.h.s. of Eq. (5.50) represents a convex combination of unitarily equivalent terms we can straightforwardly arrive at the desired Eq. (5.49) by applying joint convexity together with the invariance properties of F . $\square$

5.5 Convexity and monotonicity under the trace

Trace inequalities

Theorem 5.17 (Convex functions and positive maps under the trace)

Let $\{T_i : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C})\}$ be a collection of positive linear maps, $f : I \subseteq \mathbb{R} \rightarrow \mathbb{R}$ a function which is convex on I and $\{X_i \in \mathcal{M}_d(\mathbb{C})\}$ a collection of Hermitian operators with spectrum in I . Then

$$\operatorname{tr} \left[C f \left(\sum_i T_i(X_i) \right) \right] \leq \sum_i \operatorname{tr} [C T_i(f(X_i))] \quad (5.51)$$

holds for all positive semi-definite $C \in \mathcal{M}_{d'}(\mathbb{C})$ which commute with $f(\sum_i T_i(X_i))$ if either

1. $\sum_i T_i(\mathbb{1}) = \mathbb{1}$, or
2. $\sum_i T_i(\mathbb{1}) \leq \mathbb{1}$ and $f(0) \leq 0$.

PROOF By linearity it is sufficient to consider $C = |\psi\rangle\langle\psi|$, where ψ is a normalized eigenvector of $f(\sum_i T_i(X_i))$. Denote by $X_i = \sum_x \mu_{x,i} |x_i\rangle\langle x_i|$ the spectral decomposition of X_i . Then

$$\langle\psi|f\left(\sum_i T_i(X_i)\right)|\psi\rangle = f\left(\langle\psi|\sum_i T_i(X_i)|\psi\rangle\right) \quad (5.52)$$

$$\begin{aligned} &= f\left(\sum_{x,i} p_{x,i} \mu_{x,i}\right), \quad p_{x,i} := \langle\psi|\sum_i T_i(|x_i\rangle\langle x_i|)|\psi\rangle \\ &\leq \sum_{x,i} p_{x,i} f(\mu_{x,i}) = \sum_i \langle\psi|T_i(f(X_i))|\psi\rangle. \end{aligned} \quad (5.53)$$

Here the first equality follows from ψ being an eigenvector. To see this recall that applying a function to a Hermitian matrix means applying it to the eigenvalues (see Eq.(5.15) and the discussion of functional calculus). For the inequality in Eq.(5.53) we use convexity of f together with the fact that the $p_{x,i}$'s are positive (due to positivity of the T_i 's) and sum up to one if $\sum_i T_i(\mathbb{1}) \leq \mathbb{1}$ holds with equality. If equality does not hold we can add another term which completes the probability distribution and for which the corresponding $\mu = 0$. In this case the inequality then follows from convexity together with $f(0) \leq 0$. $\square$

The following is a simple but important consequence of the above Thm.5.17. It follows by setting $C = \mathbb{1}$ and $T_i = \lambda_i \operatorname{id}$ with λ being a probability vector:

Corollary 5.6 (Convex trace functions) *Let $f : I \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be a function which is convex on I . Then the map $X \mapsto \operatorname{tr}[f(X)]$ is convex on the set of Hermitian operators with spectrum in I .*

In order to get a similar result for function which are monotone, but not necessarily operator monotone, it is convenient to recall how functions are differentiated under the trace. If $f : I \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is a continuously differentiable function with derivative f' , then

$$\frac{\partial}{\partial x} \operatorname{tr}[f(A + xB)] \Big|_{x=x_0} = \operatorname{tr}[B f'(A + x_0 B)], \quad (5.54)$$

for all Hermitian operators $A, B \in \mathcal{M}_d(\mathbb{C})$ for which the spectrum of $A + xB$ in a neighborhood of x_0 is contained in I .⁴ With this at hand we obtain:

Proposition 5.2 (Monotone trace functions) *Let $f : I \rightarrow \mathbb{R}$ be a continuously differentiable and non-decreasing function on an open interval $I \subseteq \mathbb{R}$. Then for all Hermitian operators $A, B \in \mathcal{M}_d(\mathbb{C})$ with spectrum in I and $A \leq B$ we have that*

$$\operatorname{tr}[f(A)] \leq \operatorname{tr}[f(B)]. \quad (5.55)$$

PROOF Consider the function $g(x) := \operatorname{tr}[f(A + x\Delta)]$ with $\Delta := (B - A) \geq 0$. Following Eq.(5.54) we have $g'(x_0) = \operatorname{tr}[\Delta f'(A + x_0\Delta)]$ which is non-negative for $x_0 \geq 0$ since it is the trace of the product of two positive semi-definite operators. Thus $g(0) \leq g(1)$. $\square$

A slightly stronger result can be obtained from Weyl's monotonicity theorem which states that for any pair of Hermitian $d \times d$ matrices for which $A \leq B$, it holds that

$$\forall j : \quad \lambda_j^\downarrow(B) \geq \lambda_j^\downarrow(A) + \lambda_d^\downarrow(B - A), \quad (5.56)$$

where the $\lambda_j^\downarrow$'s are the decreasingly ordered eigenvalues. Since $\lambda_d^\downarrow(B - A) \geq 0$, one obtains immediately:

Proposition 5.3 *For any pair of Hermitian operators $A, B \in \mathcal{M}_d(\mathbb{C})$ there exists a unitary $U \in \mathcal{M}_d(\mathbb{C})$ so that for any $f : I \rightarrow \mathbb{R}$ which is non-decreasing on an interval I which includes the spectra of both A and B , we have*

$$f(A) \leq U f(B) U^\dagger. \quad (5.57)$$

adding unitaries

5.6 Operator means

⁴This is easily verified by looking at a polynomial approximation of f .

